

Operations on Sets

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A **union** of two sets, denoted by $A \cup B$, contains all elements that are in A , in B , or in both.

Intersection

An **intersection** of two sets, denoted by $A \cap B$, contains only the elements common to both sets.

$$A = \{1, 2, 3\}$$

$$B = \{2, 3, 4\}$$

Therefore,

$$A \cap B = \{2, 3\}$$

Difference of Sets

The **difference of sets**, denoted by $A - B$, contains the elements that are in A but not in B .

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

Therefore,

$$A - B = \{1, 2\}$$

Complement

The **complement** of a set, denoted by A' , contains all elements in the **universal set** that are not in A .

$$U = \{1, 2, 3, 4, 5\}$$

$$A = \{1, 3, 5\}$$

Therefore,

$$A' = \{2, 4\}$$

Examples

Example 1

Let

$$A = \{1, 2, 3, 5, 7\}$$

and

$$B = \{2, 4, 5, 6, 8\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A \cap B = \{2, 5\}$$

Example 2

Let

$$A = \{1, 2, 3, 5, 7\}$$

and

$$B = \{2, 4, 5, 6, 8\}.$$

Then

$$A - B = \{1, 3, 7\}$$

and

$$B - A = \{4, 6, 8\}.$$

Example 3

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

and

$$A = \{2, 4, 6, 8, 10\}.$$

The elements of U that are not in A are

$$A' = \{1, 3, 5, 7, 9\}.$$

Practice Exercises

Practice 1

Let $A = \{1, 2, 3, 5, 7\}$ and $B = \{2, 4, 5, 6, 8\}$.

Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

Practice 2

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

and

$$A = \{2, 4, 6, 8, 10\}.$$

Find A' .

Practice 3

Let $A = \{a, b, c, d, e\}$ and $B = \{c, d, e, f, g\}$.

Find $A \cup B$, $A \cap B$, and $A - B$.

Practice 4

Let $A = \{2, 4, 6, 8\}$ and $B = \{1, 2, 3, 4, 5\}$.
Find $(A \cup B)$ and $(A \cap B)$.

Practice 5

Let

$$U = \{a, b, c, d, e, f, g, h\}$$

and

$$C = \{b, d, f, h\}.$$

Find C' .

Practice 6

Let $M = \{3, 6, 9, 12, 15\}$ and $N = \{6, 12, 18, 24\}$.

Find $M \cup N$, $M \cap N$, and $M - N$.

Practice 7

Let $P = \{x, y, z\}$ and $Q = \{y, z, w, v\}$.

Find $P \cup Q$, $P \cap Q$, and $Q - P$.

Practice 8

Let

$$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

and

$$R = \{1, 3, 5, 7, 9\}.$$

Find R' and $R \cup \{0\}$.

Practice 9

Let $A = \{2, 3, 5, 7, 11\}$ and $B = \{5, 7, 13, 17\}$.

Find $A \cup B$, $A \cap B$, and $B - A$.

Activities

Activity 1

Let $A = \{3, 5, 7, 9, 11\}$ and $B = \{1, 3, 5, 8, 10\}$.

Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

Activity 2

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

and

$$A = \{1, 3, 5, 7, 9, 11\}.$$

Find A' .

Activity 3

Let $A = \{m, n, o, p, q\}$ and $B = \{o, p, q, r, s\}$.

Find $A \cup B$, $A \cap B$, and $A - B$.

Activity 4

Let $A = \{5, 10, 15, 20\}$ and $B = \{10, 15, 25, 30\}$.

Find $(A \cup B)$ and $(A \cap B)$.

Activity 5

Let

$$U = \{a, b, c, d, e, f, g, h, i, j\}$$

and

$$C = \{a, c, e, g, i\}.$$

Find C' .

Activity 6

Let $M = \{4, 8, 12, 16, 20\}$ and $N = \{8, 16, 24, 32\}$.

Find $M \cup N$, $M \cap N$, and $M - N$.

Activity 7

Let $P = \{k, l, m\}$ and $Q = \{m, n, o, l\}$.

Find $P \cup Q$, $P \cap Q$, and $Q - P$.

Activity 8

Let

$$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

and

$$R = \{0, 2, 4, 6, 8\}.$$

Find R' and $R \cup \{9\}$.

Thank you!